

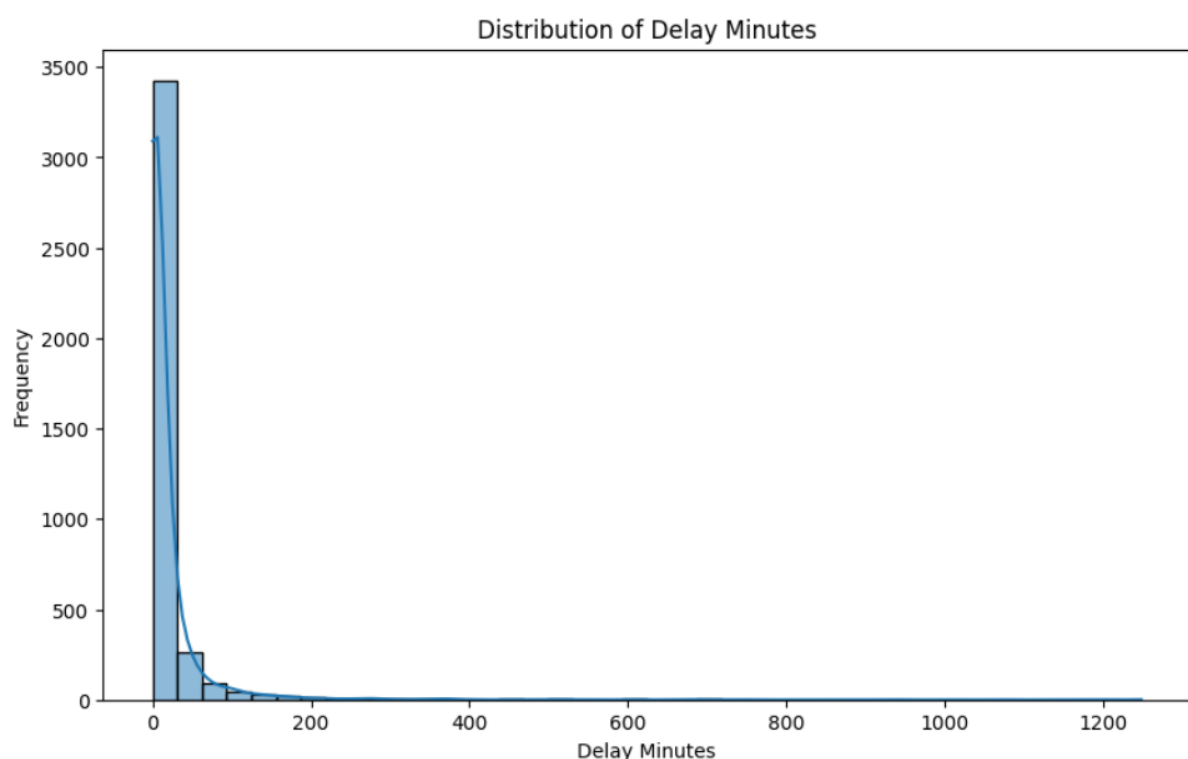
Exploratory Data Analysis

(RR w2073057)

Distribution of Delay Minutes:

The distribution of delay minutes was examined to understand how delays are spread across incidents. The histogram shows that the data is highly right-skewed. Most incidents generate relatively small delays, while a small number of incidents result in very large delay minutes.

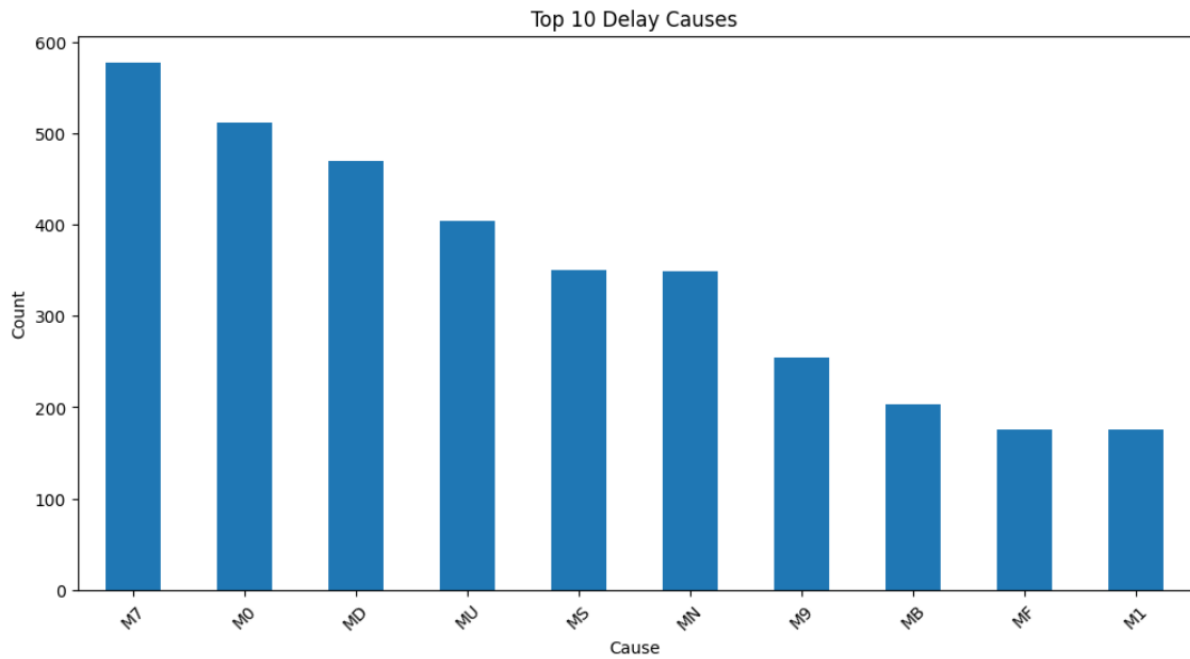
This pattern is typical in railway operations, where most disruptions are minor but occasional incidents can cause severe network disruption.



Delay Causes:

The most common delay causes were analysed using a bar chart showing the top ten cause codes in the dataset. Although the causes are recorded as abbreviated codes, the results still highlight which categories of operational issues occur most frequently.

A small number of delays causes account for a large proportion of incidents, suggesting that certain operational problems occur repeatedly within the network.

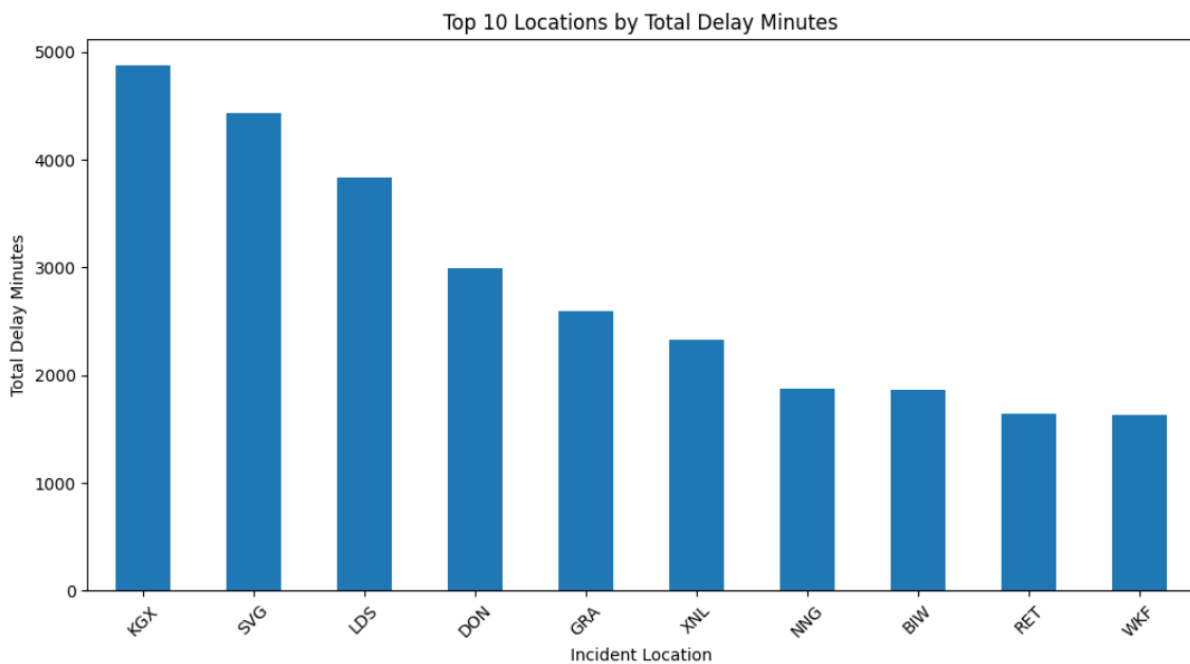


Incident Locations:

Total delay minutes were aggregated by incident location to identify where disruptions generate the largest operational impact.

The results show that several key locations account for a large share of total delay minutes. These locations are likely major operational hubs where disruptions can affect multiple services simultaneously.

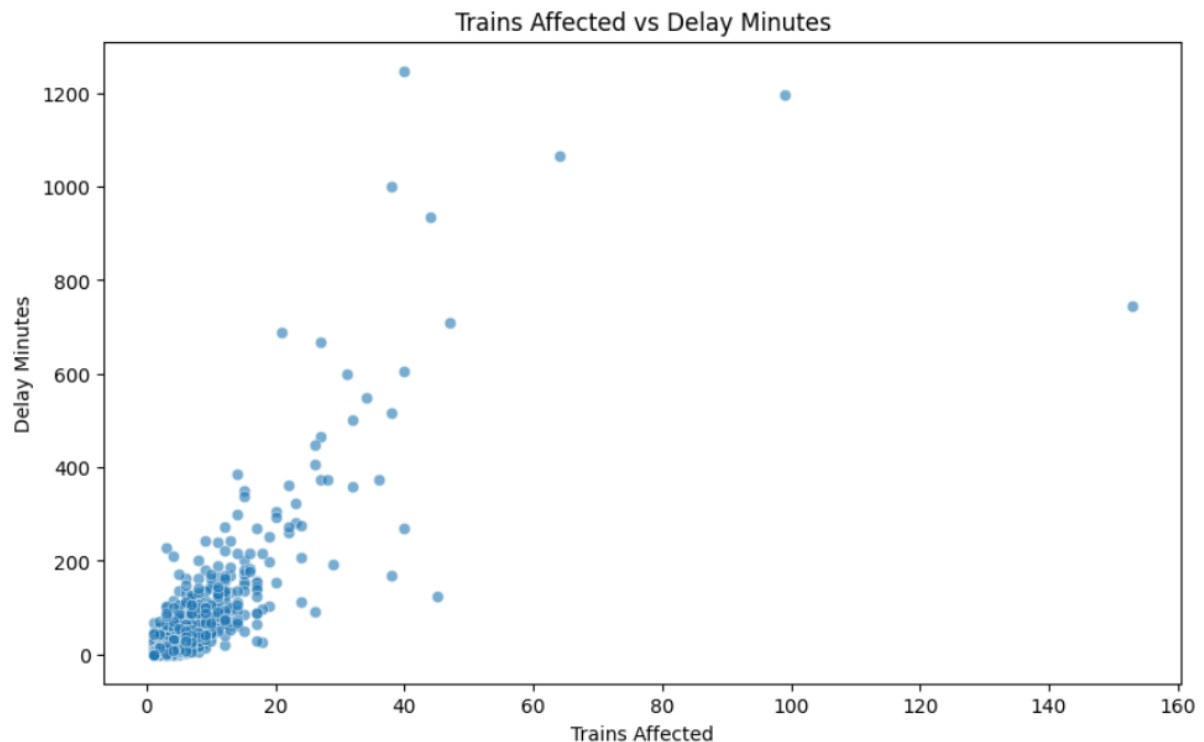
Because the location codes are abbreviated, the analysis focuses on identifying relative disruption levels rather than interpreting individual station names.



Trains Affected Vs Delay Minutes:

A scatter plot was used to explore the relationship between the number of trains affected and delay minutes.

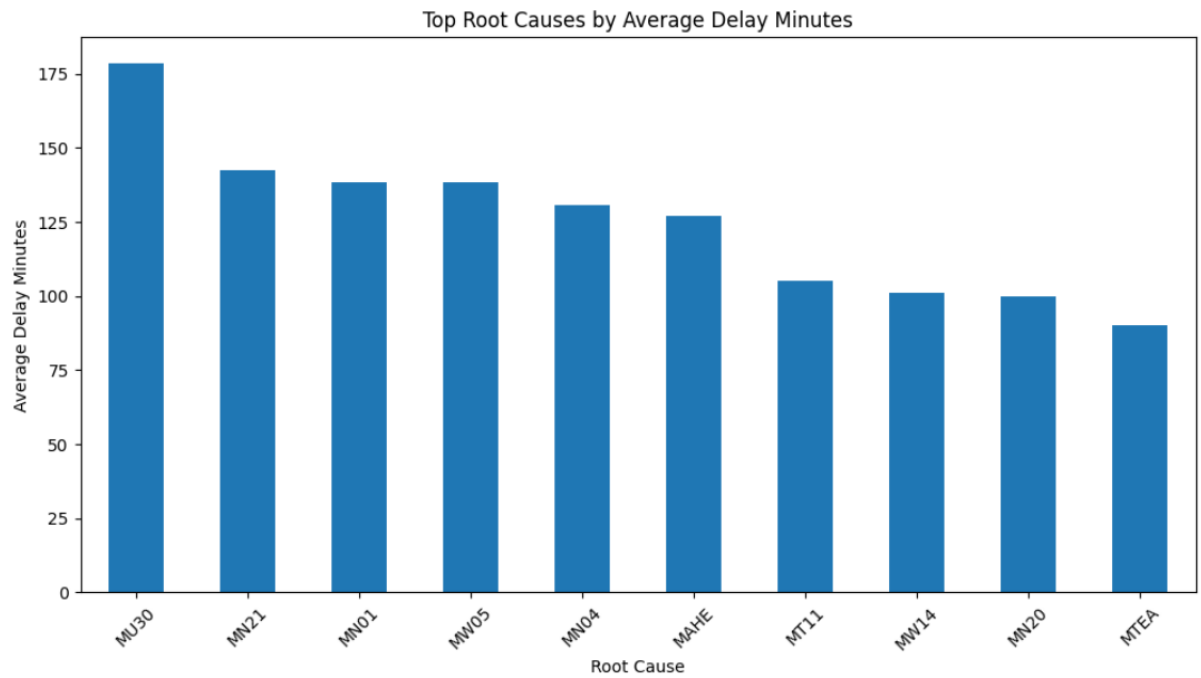
The results show a general positive relationship between the two variables. Incidents affecting a larger number of trains tend to generate higher delay minutes. This reflects how disruptions can propagate through railway networks, where delays affecting multiple trains can significantly increase the overall operational impact.



Root Cause Severity:

To understand which incident types produce the most severe disruptions, the average delay minutes were calculated for each root cause category.

The results show that certain root causes are associated with significantly higher average delay minutes. This suggests that although some incidents may occur less frequently, they can generate disproportionately large operational disruption.

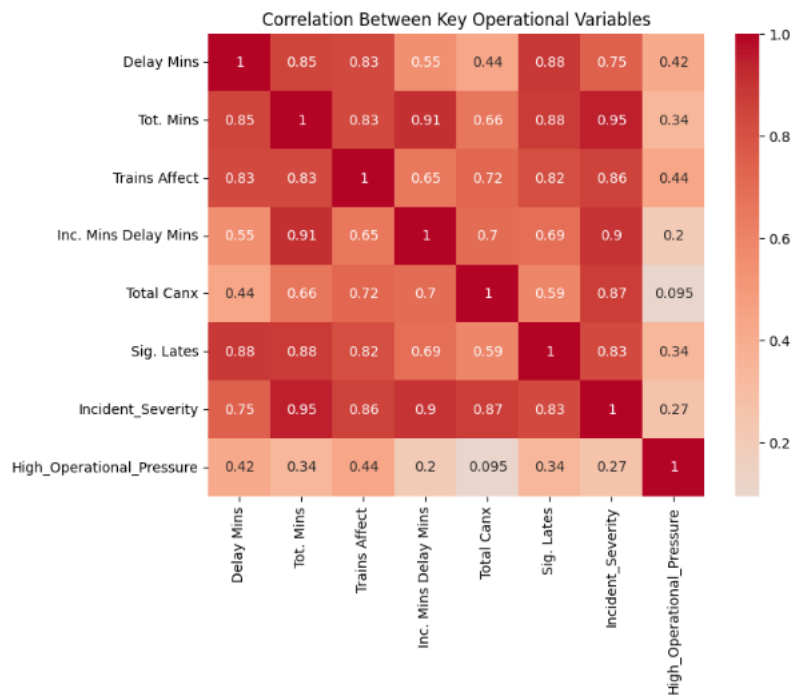


Correlation Analysis:

A correlation heatmap was used to examine relationships between key operational variables such as delay minutes, total delay minutes, trains affected and incident severity.

The heatmap shows strong positive correlations between several disruption metrics. For example, delay minutes and total delay minutes are strongly related, which is expected since severe incidents generate larger network delays. Trains affected also shows a strong relationship with delay minutes, suggesting that incidents impacting multiple services produce greater disruption.

This analysis helped identify which variables are most strongly related to operational pressure and therefore useful for predictive modelling.



Distribution of Operational Pressure:

A binary target variable called High Operational Pressure was created to classify incidents into two categories: normal operational pressure and high operational pressure.

The distribution plot shows how incidents are split between these two classes. High operational pressure incidents represent the most severe disruptions in the dataset and are the main focus of the predictive modelling stage.

```
High_Operational_Pressure
0    2965
1     960
Name: count, dtype: int64
Threshold used: 26.0
```

Feature Engineering:

Before training the models, several additional features were created to capture the severity of incidents more effectively. These engineered features included measures such as incident severity and delay per train, which combine multiple operational indicators into single variables.

A threshold based on delay metrics was also used to define the High Operational Pressure target variable. This allowed the modelling process to focus specifically on predicting severe disruption events.

One – Hot Encoding:

Some variables in the dataset, such as root cause and incident location, are categorical rather than numeric. Machine learning models cannot work directly with text categories.

To address this, one-hot encoding was used to convert categorical variables into binary indicator variables. Each category is represented as a separate column indicating whether that category is present for a particular incident.

This allows the machine learning models to incorporate categorical operational factors into the prediction process.

```
1 #one hot encode categorical variables
2 X = pd.get_dummies(X, drop_first=True)
3
4 print(X.shape)
5 print(X.head())
```

```
... (3925, 946)
   Delay Mins  Inc. Mins Delay Mins  Trains Affect  EQ Mins  Capes  Pines  \
0         0.0         0.0         8.0           1         8.0     0.0     0.0
1         2.0         0.0         0.0           1         2.0     0.0     0.0
2         0.0         0.0         6.0           1         6.0     0.0     0.0
3         3.0         0.0         0.0           1         3.0     0.0     0.0
4        181.0        143.0        504.0          9        504.0     2.0     1.0

   Calvs  FTS  Full Canx  Part Canx  ... Equip._UNKNOWN Equip._Undefined  \
0     0.0   0     0.0     0.0  ...      False             False
1     0.0   0     0.0     0.0  ...      False             False
2     0.0   0     0.0     0.0  ...      False             False
3     0.0   0     0.0     0.0  ...      False             False
4     0.0   0     3.0     0.0  ...      False             False

   Equip._Unknown Equip._VARIOUS KPI_701D KPI_750 NR KPI_Fleet  \
0         False         False      True   False      True
1         False         False      True   False      False
2         False         False      True   False      True
3         False         False      True   False      True
4         False         False      True   False      True

   Region_Depot  Region_IC225  Region_Other
0         False         False      False
1         True          False      False
2         False         False      False
3         False         False      False
4         False          True      False

[5 rows x 946 columns]
```

Predictive Modelling:

Three machine learning models were trained to predict high operational pressure incidents:

- Decision Tree
- Logistic Regression
- Random Forest

These models were chosen because they are commonly used for classification problems and provide different approaches to identifying patterns in the data.

Decision Tree Model:

The Decision Tree model provides an interpretable structure showing how different variables contribute to predicting operational pressure.

The model identifies incident severity as the most important predictor, followed by variables such as delay minutes and delay per train. These features represent the scale and impact of operational disruptions.

The Decision Tree achieved an accuracy of approximately 98.6%, indicating strong predictive performance.

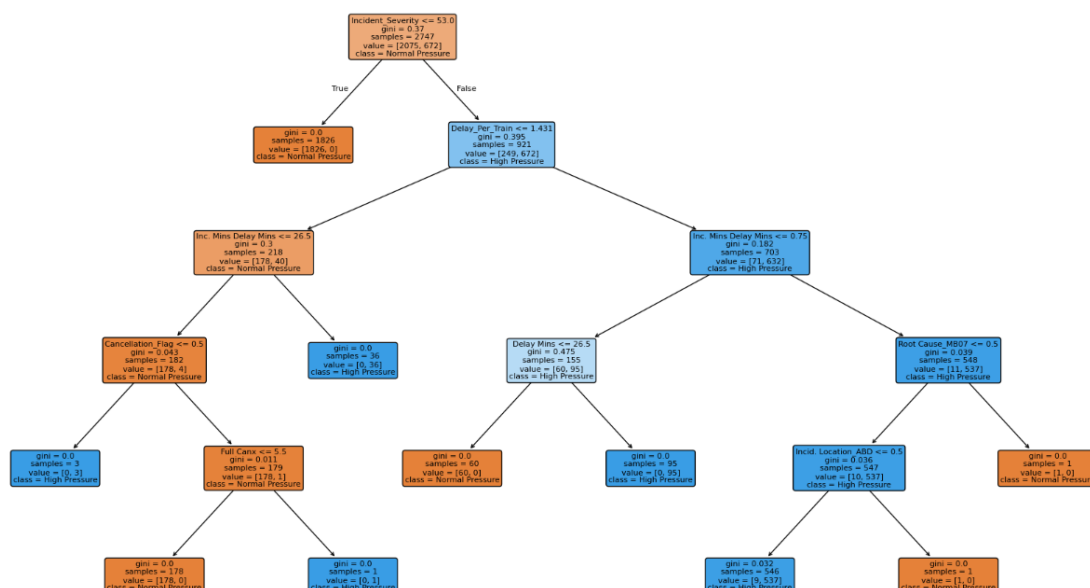
Decision Tree Accuracy: 0.9864176570458404

Decision Tree Confusion Matrix:

```
[[878 12]
 [ 4 284]]
```

Decision Tree Classification Report:

	precision	recall	f1-score	support
0	1.00	0.99	0.99	890
1	0.96	0.99	0.97	288
accuracy			0.99	1178
macro avg	0.98	0.99	0.98	1178
weighted avg	0.99	0.99	0.99	1178



Overfitting Check:

To ensure the model was not overfitting the training data, the training accuracy and testing accuracy were compared.

The training accuracy was approximately 99.7%, while the testing accuracy was approximately 98.6%. Since the two values are very close, this suggests that the model generalises well and is not significantly overfitting.

Training Accuracy: 0.9967236985882694
Testing Accuracy: 0.9864176570458404

Model Comparison:

To evaluate the robustness of the results, two additional models were trained: Logistic Regression and Random Forest.

Logistic Regression achieved the highest accuracy on the test data, while Random Forest also performed extremely well. The strong performance across multiple models suggests that the dataset contains clear patterns that allow severe disruptions to be predicted effectively.

```
Logistic Regression Accuracy: 1.0
```

Logistic Regression Confusion Matrix:

```
[[890  0]
 [ 0 288]]
```

Logistic Regression Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	890
1	1.00	1.00	1.00	288
accuracy			1.00	1178
macro avg	1.00	1.00	1.00	1178
weighted avg	1.00	1.00	1.00	1178

```
Random Forest Accuracy: 0.9923599328882852
```

Random Forest Confusion Matrix:

```
[[887  3]
 [ 6 282]]
```

Random Forest Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	890
1	0.99	0.98	0.98	288
accuracy			0.99	1178
macro avg	0.99	0.99	0.99	1178
weighted avg	0.99	0.99	0.99	1178

Limitations of the Model:

Although the predictive models achieved high accuracy, several limitations should be acknowledged.

First, many variables in the dataset represent different measures of the same disruption events. For example, delay minutes, total delay minutes and incident severity are strongly correlated. This may make the prediction task easier for the models and could partly explain the very high accuracy scores.

Second, the dataset contains abbreviated operational codes for causes and locations. While these codes allow patterns to be identified, their meanings are not directly interpretable without additional operational documentation.

Finally, the target variable for high operational pressure was created using a threshold derived from delay metrics. While this provides a practical way to classify severe incidents, the definition of operational pressure may vary depending on the operational context of the railway network.

Future work could incorporate additional contextual datasets, such as infrastructure data or operational planning information, to improve the predictive modelling framework

